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Batch – May 2026

Uber/Ola Fare Estimator Using Linear Regression

Introduction

Ride-sharing services such as Uber and Ola have become an important part of modern transportation. Estimating the fare of a ride before booking helps customers plan their travel expenses and assists service providers in maintaining transparency. Machine Learning techniques can be used to predict ride fares based on factors such as travel distance, estimated travel time, and traffic conditions.

The main objective of this project is to develop an Uber/Ola Fare Estimator using the Linear Regression algorithm. The model analyzes historical ride data and predicts the estimated fare for a given journey. Linear Regression is one of the most widely used supervised machine learning algorithms for predicting continuous numerical values and is suitable for fare prediction problems.

The project uses the Uber Fare Prediction dataset obtained from Kaggle. Various preprocessing techniques are applied to clean the data and generate meaningful features such as distance travelled. The trained model can then estimate the fare for new rides based on user-provided inputs.

Problem Statement

The main objective of the "Uber/Ola Fare Estimator" is to implement a machine learning model that predicts the fare amount of a ride using Linear Regression.

The fare amount is considered the target variable (output), while the features include distance travelled, estimated travel time, and traffic level. The system aims to provide users with an estimated fare before booking a ride.

The problem statement can be defined as follows:

"Given a dataset containing historical Uber ride information, develop a machine learning model capable of predicting ride fares based on travel distance, estimated duration, and traffic conditions. The model should be trained using historical data and evaluated on unseen data to determine its prediction performance."

The project also demonstrates how machine learning can be applied to transportation systems for cost estimation and decision-making.

Results and Discussion

The dataset was cleaned by removing missing values and invalid records. Distance between pickup and drop-off locations was calculated using geographical coordinates. Additional features such as estimated travel time and traffic level were generated to improve the prediction capability of the model.

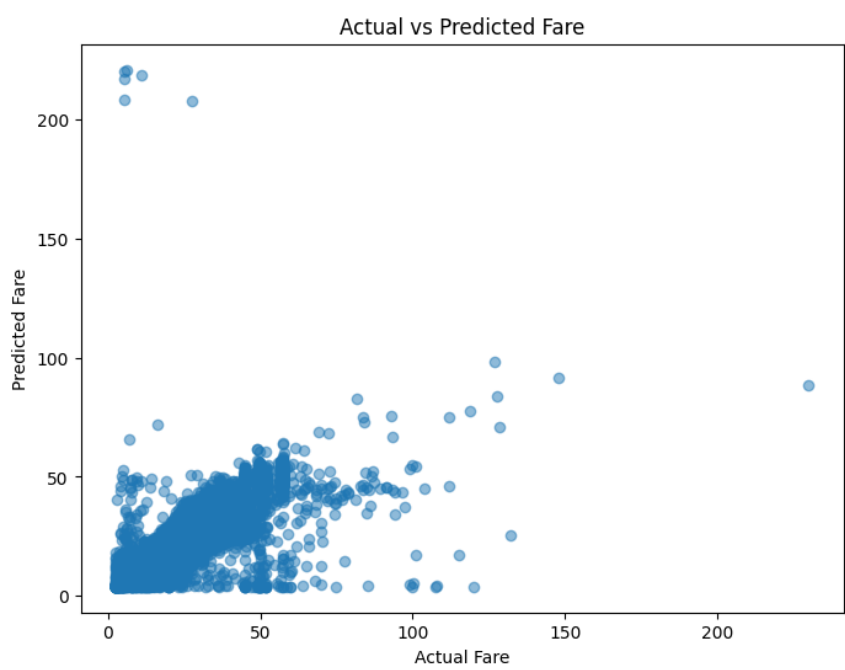
The Linear Regression model was trained using the processed dataset and evaluated using performance metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 Score.

The results indicate that ride fare is strongly influenced by travel distance and estimated travel time. Higher traffic levels generally increase the estimated fare due to longer travel durations. The trained model successfully learned the relationship between these factors and provided realistic fare predictions for new ride requests.

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... Original Dataset Shape: (200000, 9)
Shape After Cleaning: (199269, 9)

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MODEL PERFORMANCE
=====
MAE : 2.44
MSE : 29.42
RMSE : 5.42
R2 Score: 0.6944

Sample Predictions:
Actual Fare Predicted Fare
0          5.3      7.160126
1          6.5      5.141781
2         12.5     18.665565
3          6.9     10.103094
4         15.7     21.138587
5          6.5      8.391123
6          4.5      6.711910
7         14.1     14.549543
8          8.5      8.644832
9         11.0     10.033600
```



Conclusion

In this project, a machine learning-based Uber/Ola Fare Estimator was developed using the Linear Regression algorithm. The model was trained on historical Uber ride data and used distance, estimated travel time, and traffic conditions as the primary features for prediction.

The developed system demonstrates how machine learning techniques can be applied in the transportation sector to estimate ride fares accurately. Linear Regression proved to be an effective algorithm for understanding the relationship between fare amount and ride characteristics.

The project can be further improved by integrating real-time Google Maps services to automatically fetch route distance, travel duration, and live traffic information. Future enhancements may also include advanced machine learning algorithms such as Random Forest Regression and XGBoost for improved prediction accuracy.

Overall, the project provides a practical solution for ride fare estimation and showcases the application of machine learning in real-world transportation systems.

Reference

1. Uber Fare Prediction Dataset, Kaggle.
2. Scikit-Learn Documentation for Linear Regression.
3. Pandas Documentation.
4. NumPy Documentation.
5. Google Maps Platform Documentation.